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Article

BEAST DB: Grand-Canonical Database of Electrocatalyst Properties

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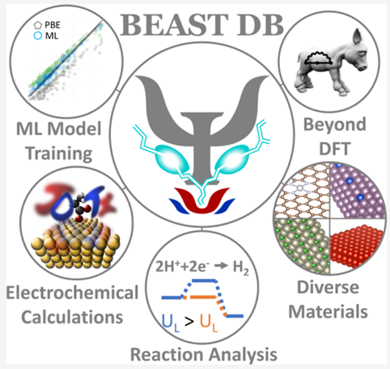
ABSTRACT: We present BEAST DB, an open-source database comprised of ab initio electrochemical data computed using grand-canonical density functional theory in implicit solvent at consistent calculation parameters. The database contains over 20,000 surface calculations and covers a broad set of heterogeneous catalyst materials and electrochemical reactions. Calculations were performed at self-consistent fixed potential as well as constant charge to facilitate comparisons to the computational hydrogen electrode. This article presents common use cases of the database to rationalize trends in catalyst activity, screen catalyst material spaces, understand elementary mechanistic steps, analyze the electronic structure, and train machine learning models to predict higher fidelity properties. Users can interact graphically with the database by querying for individual calculations to gain a granular understanding of reaction steps or by querying for an entire reaction pathway on a given material using an interactive reaction pathway tool. BEAST DB will be periodically updated, with planned future updates to include advanced electronic structure data, surface speciation studies, and greater reaction coverage.



1. INTRODUCTION

Electrocatalysis is a cornerstone technology that underlies numerous renewable energy and chemical processes crucial for realizing a decarbonized economy.[1] Electrocatalysts are expected to play a central role in the production of sustainable chemicals, such as the production of green hydrogen, providing both sustainable fuel, feedstocks to other chemical processes, and long-term energy storage solutions.[2] Additionally, electrocatalysts facilitate the generation of electricity by fuel cells, enhancing their efficiency and minimizing the environmental impact of converting chemical energy back into electrical energy.[3] Beyond providing energy solutions, electrocatalysis enables circularization of the chemical industry through sustainable generation of value-added chemicals. This includes the electrochemical reduction of carbon dioxide into carbon-neutral products and the production of ammonia for fertilizers from alternative nitrogen sources.[4,5]

Advances in catalyst design and performance therefore hold significant potential for reducing our reliance on fossil fuels and minimizing environmental impact. However, gaining an understanding of electrocatalytic processes is partly hindered by the wide variety of conditions (e.g., potential ranges, catalyst composition, and pH) under which experiments are performed. This diversity makes it challenging to determine the individual impact of these factors on performance.

Computational approaches offer a valuable means to rationalize these choices by providing mechanistic insight and enabling identification of trends across data more quickly than experiment.[6−8] However, they also face the complexity of simulating a wide variety of experimental conditions, which can complicate such comparisons. Data science offers promise in addressing the challenges posed by the diverse conditions encountered in electrocatalysis research, providing a comprehensive framework for analysis of electrochemical performance and development of standardized protocols and methodologies. In the materials domain, the ability to readily access material properties, including relative stability, structure, and electronic structure, through a simple database query has significantly benefited researchers. This has been facilitated through large public databases of consistently computed properties such as the Materials Project,[9] NOMAD,[10] OQMD,[11] AFLOW,[12] and AiiDA.[13] While the largest databases use density functional theory (DFT) as their underlying theory, the C2DB provides systematic beyond-DFT properties for 2D materials and Schmidt and Thygesen have created a database of roughly 200 random phase approximation (RPA) adsorption energies on monometallic metals.[14,15] These resources enable researchers to develop novel approaches for discovering functional materials and use the trends within the compiled data to guide further research.

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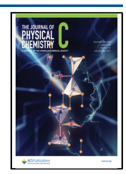

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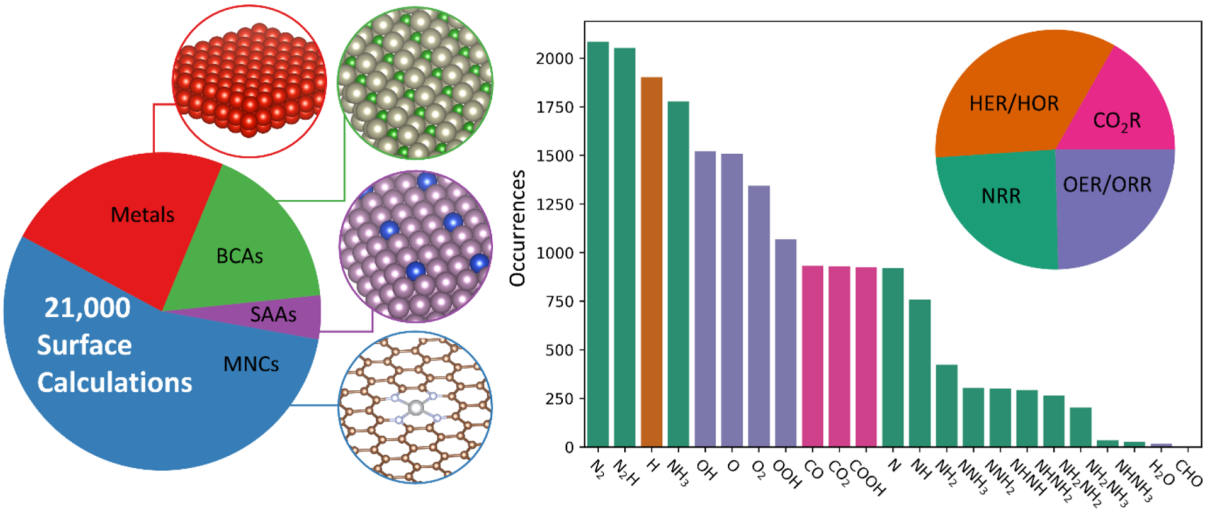

Figure 1. Database overview. (Left) The database is comprised of four main material classes: graphene-embedded single-atom catalysts (MNCs, 55%), monometallic flat and stepped metals (23%), binary covalent alloys (BCAs, 17%), and bimetallic single-atom alloys (SAAs, 5%). (Right) The distributions of reaction intermediate calculations are shown for the four main reactions: hydrogen evolution/reduction (HER/HOR), nitrogen reduction (NRR), oxygen evolution/reduction (OER/ORR), and CO_2 reduction (CO2R).

Materials databases have paved the way for accessing computed properties, but catalysis databases face unique challenges in capturing the complexity of catalytic systems. In the catalysis space, multiple databases exist with different foci: machine-learning DFT structural relaxations,[16] collating data on different catalysts and chemistries for heterogeneous catalysis based on the existing literature,[17] and developing scaling relations over relatively large combinatoric spaces of electrocatalysts.[18] While these databases have provided many insights within their respective areas of focus, they share common characteristics that limit their ability to inform electrocatalysis studies; they (1) either neglect the applied potential and solvent effects or treat them at the computational hydrogen electrode (CHE) level,[19] (2) compute only adsorption energies or reaction energetics, and (3) use DFT as their most accurate level of theory. Additionally, only Open Catalyst Project uses a uniform set of calculation parameters for all the calculations tabulated in the database.

The compiled electrocatalyst properties tabulated within current databases to date[16,18] have used the CHE[19] (with some exceptions in Catalysis-hub) to approximate the effect of the applied potential on reaction energetics. The CHE model is the most widely used approach to include the effect of potential in electrochemical processes primarily because it only requires that one applies a simple potential-dependent, postprocessing correction that shifts the energetics of proton-coupled electron transfer (PCET) steps to otherwise standard uncharged DFT calculations.[20] Since its conception, CHE has been successfully used to rationalize observed trends in electrochemical reactions and design electrocatalysts that facilitate desired electrochemical conversions.[21,22] While the CHE approach has been successful, it is inherently limited in modeling the complete electrochemical environment.[23] For instance, it does not describe the effect of applied potential on activation barriers, the geometries of reacting species, or the energies of any non-PCET step.

In contrast, grand-canonical density functional theory (GC-DFT) provides a rigorous theoretical framework to explicitly account for the influence of the electrochemical potential and charged species present in the electrolyte on the free energies of reaction steps.[24,25] In GC-DFT, the electrostatic potential contributions from both the electrode and electrolyte are explicitly incorporated into the DFT calculations, allowing the electrode electronic charge and the corresponding induced electrolyte ionic charge to respond self-consistently to the applied potential. This enables modeling of electrochemical interface processes beyond PCETs and potential-dependent binding of reaction intermediates while including the effects of electrolyte solvation. By considering these additional effects, GC-DFT provides a more comprehensive and fundamental description of how the reaction energy landscape shifts with electrode potential. The database introduced here leverages the realistic treatment of the GC-DFT approach to evaluate electrochemical reactions across a wide range of operating conditions.

To make clear comparisons between various electrocatalysts under different operating conditions, we have constructed BEAST DB: the Beyond-DFT Electrochemistry with Accelerated and Solvated Techniques database. BEAST DB tabulates applied potential and continuum solvation effects computed using GC-DFT, thus providing quantities that are important for rationalizing computed adsorption energies as a function of potential (e.g., charges on molecule and active site atoms) and facilitating the comparison of calculations at equivalent calculation parameters. Ultimately, BEAST DB will also provide adsorption energies and electronic structure at a beyond-DFT level for the most promising catalysts and chemistries. Herein, we introduce BEAST DB and provide an overview of its contents and methods.

Throughout, we include example studies of specific systems to illustrate the capabilities afforded by BEAST DB. We compare the CHE with solvation and GC-DFT methods, investigating and contrasting their impacts on thermodynamic activity descriptors and reaction intermediate scaling relationships. Furthermore, we explore electronic structure variations under applied potential and their implications for predicted reaction energetics. We discuss GW and random-phase approximation (RPA) energy level calculations of select catalysts to assess how beyond-DFT exchange-correlation effects influence computed reaction energetics and our plans to

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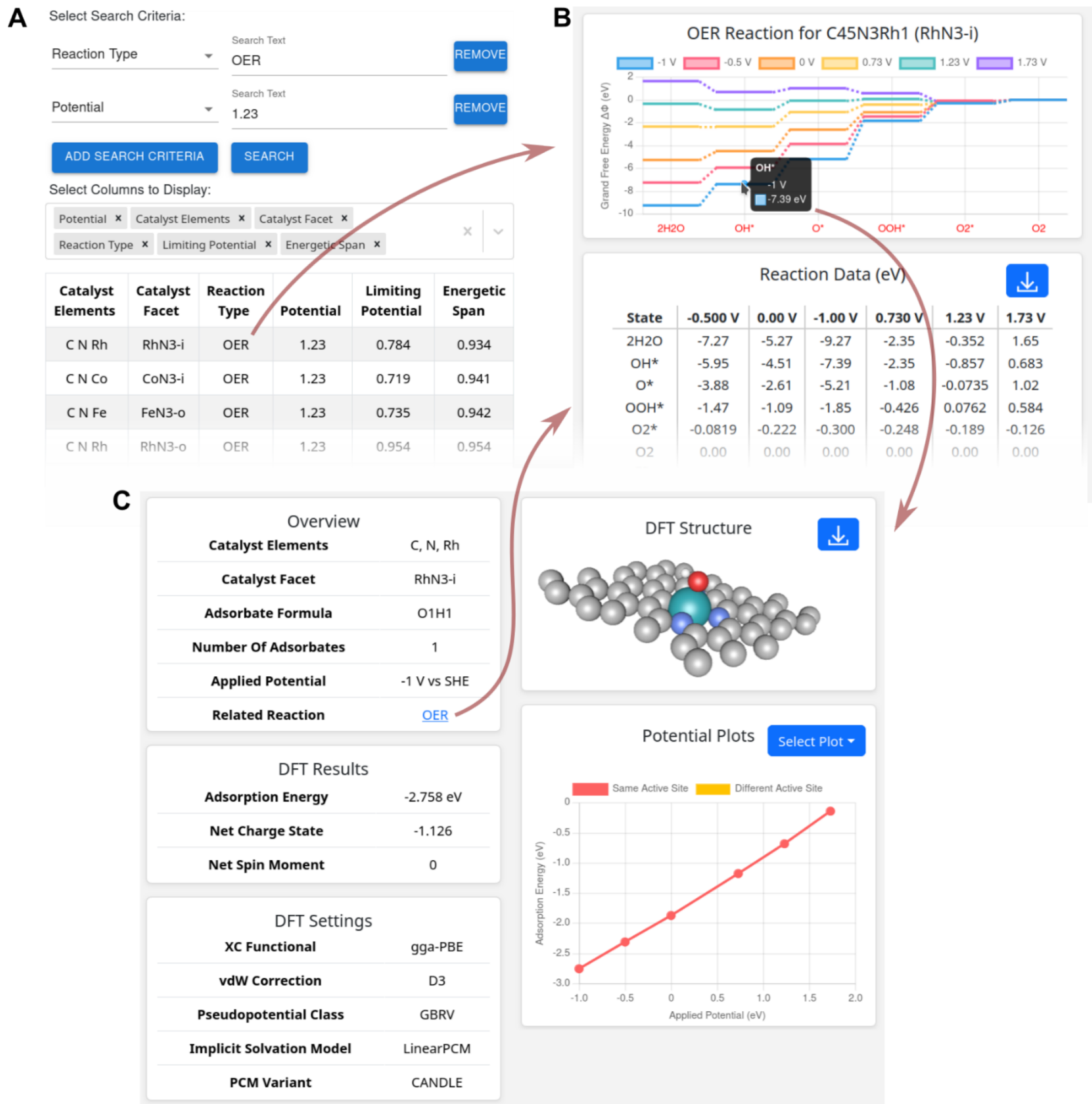

Figure 2. User interface. (A) Reaction search based on catalyst elements, reaction type, potential, and activity descriptors, shown here for OER at 1.23 V with sorting based on the energetic span. Each row links to panel (B), a detailed view of the reaction pathway as a function of potential. Each intermediate in the pathways links to panel (C), a detailed view of a specific catalyst with adsorbate calculation, including calculation details and plots of the structure and potential dependence of properties such as energy and charge (selectable). BEAST DB also allows searching directly for the adsorbates on catalysts (not shown here), linking to panel (C), which in turn links to corresponding reaction pathways.

extend these calculations to the most promising catalysts in the database. Finally, we discuss how a database of properties computed with high-level theories such as GW and RPA enables machine learning (ML)-based delta-learning to achieve beyond-DFT accuracy at the computational cost of standard DFT.

1.1. Database Overview. BEAST DB is designed to study the reaction pathways of several electrochemical redox processes by explicitly accounting for the impact of solvation and electrode potential.[25,26] The database contains individual GC-DFT calculations of adsorbates on catalyst surfaces as well as the adsorbate molecules and surfaces separately to facilitate binding energy calculations. Each calculation is counted as a combination of a material, surface facet, adsorbate, active site, and potential. BEAST DB currently compiles the results of >21,000 calculations, spanning four unique material spaces and four distinct electrochemical reaction pathways (Figure 1). The database entries include single metal atom nitrogen-doped graphene catalysts (MNCs), flat and stepped pure metal surfaces, binary covalent alloys (BCAs), and bimetallic single-atom alloys (SAAs). The database currently includes adsorbates for the study of CO_2 reduction (CO2R), oxygen evolution/reduction (OER/ORR), hydrogen evolution/reduction (HER/HOR), and nitrogen reduction (NRR) pathways, although additional adsorbates relevant to the study of other reactions will be incorporated in subsequent releases. The coverage of reactions and their corresponding intermediates are shown in Figure 1, which shows that HER/HOR is the most common reaction in the data set followed by NRR, OER/ORR, and CO2R. The MNC data set present in the database was initialized in a high spin state with a +5 magnetic moment on the single metal atom, and the magnetization was allowed to relax. MNC surfaces were modeled with pyridinic and pyrrolic nitrogen configurations (example structures are in Section 3 of the Supporting Information), and N1−N4 defects were studied for each metal and nitrogen configuration.

The database interface allows users to search for reactions, catalysts, and adsorbates in several ways. Figure 2 shows a typical workflow, starting from a search of catalysts by reaction type, with OER shown as an example (Figure 2A). This search can be filtered by any combination of catalyst elements and applied potential and thermodynamic descriptors such as limiting potential and energetic span, as well as sorted by any of these fields. The example shows the lowest energetic span OER catalysts at the selected potential. Each row links to a detailed reaction view (Figure 2B), showing the reaction pathways as a function of potential, along with a table of energies for all steps that can be downloaded for further external analysis. Each point on the pathway links to a detailed view of the catalyst + adsorbate configuration (Figure 2C), showing the structure, calculation details, and resulting properties for that configuration. Importantly, this interface includes a potential-dependence plot that groups together all calculations available for a specific adsorbate on a surface, regardless of the electrode potential and adsorption site, e.g., atop, bridge, hollow site, etc. This facilitates viewing the influence of the electrode potential as well as variations with respect to the adsorbed site on properties including adsorption energies, total charges, and atomic charges of the active site and adsorbate.[27] This also allows for understanding how charge transfer occurs during a reaction, how charge transfer changes as a function of potential, and how the observed charge transfer corresponds to observed trends in the reaction energetics as a function of potential. BEAST DB also allows searching directly for adsorbate configurations, leading to the interface in Figure 2C, which then links to all corresponding reaction pathways. Overall, the interface allows finding optimal catalysts and adsorbate configurations of interest and analyzing the impact of potential on all properties relevant for electrocatalytic reactions.

BEAST DB facilitates comparison of GC-DFT to the CHE applied to canonical DFT calculations. Figure 3 compares the binding energies calculated by GC-DFT at a self-consistent bias vs those calculated by canonical DFT under the same solvation.

$$\Delta\Phi_{GC-DFT}[U] = \Phi_{adsorbed}[U] - (\Phi_{surface}[U] + \Phi_{molecules}[U]) \tag{1}$$

$$\Delta G_{CHE}[U] = G_{adsorbed} - (G_{surface} + G_{molecules}) - N_e U \tag{2}$$

GC-DFT self-consistently calculates energies as a function of the electrode potential, while the CHE binding energy calculations are done at the potential of zero charge and include the electrode potential dependence as a linear N_eU term, where N_e is the number of formal electron transfers in a binding event and U is the electrode potential vs the standard hydrogen electrode (SHE). At 0 V vs SHE, the agreement between CHE and GC-DFT is the greatest, with a mean difference of 0.01 eV and a standard deviation of 0.74 eV.

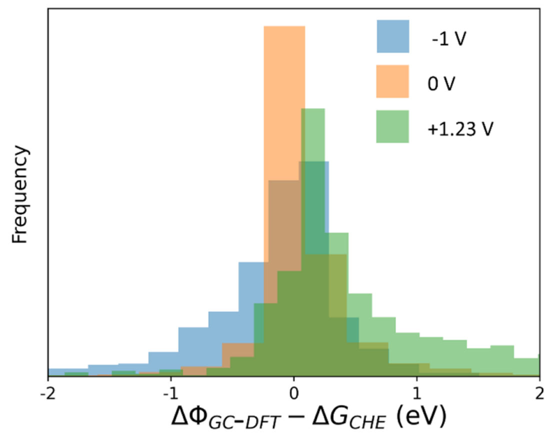

Figure 3. Global CHE comparison. All voltages are with respect to SHE. Binding energies were calculated at self-consistent applied bias with GC-DFT from which binding energies calculated with CHE at neutral charge were subtracted. GC-DFT and CHE show better agreement the closer they are to 0 V vs SHE. Equations 1 and 2 show how binding energies are calculated.

However, as the applied potential moves further from 0 V vs SHE, so does the difference in calculated binding energies. At −1.00 and +1.23 V, the mean differences are −0.19 and 0.52 eV, respectively. We note that the adsorbates present at each of these biases are different, with calculations at +1.23 V composed of primarily OER/ORR species and calculations at −1.00 V composed of HER and NRR adsorbates. This difference in reaction intermediates may lead to the different directions in which the mean differences change, as it has been shown that some intermediate reaction steps display opposite sensitivities to applied potential.[28,29] The methods used to generate this data set have demonstrated previous qualitative agreement with CHE for material activity ordering.[30] Furthermore, Sections 1 and 4 in the Supporting Information show comparisons of our methods to well-established HER binding calculations and demonstrate that the majority of the calculations in the data set exhibit small structural differences from canonical calculations, although there are instances of important structural change only captured by GC-DFT.

Although GC-DFT rigorously models the applied electrode potential, it is often still reliant on semilocal DFT functionals, which have been shown to be fundamentally limited in their accuracy toward predicting adsorption energies and relative favorability of binding sites, which is difficult to remedy without more advanced theory.[31] BEAST DB is primarily populated by results calculated using GC-DFT. However, to quantify errors associated with more computationally expedient approaches, we are also developing workflows that facilitate the significantly more expensive beyond-DFT GW and RPA calculations. In an upcoming release of BEAST DB, we will provide GW electronic structures and RPA total energies for a subset of the data consisting of transition metal surfaces (Ag, Cu, Ni, Pt, and Ru) and a 2D catalyst (FeN_4@G) with small adsorbates important to the above electrochemical reactions (CO, H_2, N_2, and O_2). Currently, these calculations are performed for surfaces in vacuum because methodological developments are still needed to facilitate the calculation of RPA total energies with solvation and applied potential.

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2. METHODS

2.1. Density Functional Theory. The GC-DFT results in the database were calculated using the charge-asymmetric nonlocally determined local-electric (CANDLE) solvation model.[26] CANDLE has been shown to accurately account for solvation effects, accurately predict the pK_a value of several molecules, better treat charge asymmetry between cations and anions, and allow for overall cell neutrality to be maintained when surfaces charge in GC-DFT calculations.

Within the grand-canonical ensemble at a specified electrochemical potential vs vacuum, μ, the total grand free energy of a system, $\Phi[\mu]$, can be written as eq 1:

$$\Phi[\mu] = F_{DFT} - \mu N + E_{ZPE} + \int_0^T C_p dT - TS \tag{3}$$

where F_{DFT} is the DFT Helmholtz energy and equal to the DFT total energy, N is the total number of electrons in the system, C_p is the system heat capacity at constant pressure, T is the temperature, and S is the entropy. The DFT-computed Helmholtz energy is defined as $F_{DFT} = E_{DFT} - TS_{occ}$ for these systems because variable partial orbital occupations were used, where S_{occ} is the entropy introduced due to the use of broadening, as is common for calculations involving metals. E_{DFT} is the DFT internal energy. We note that the energies reported in BEAST DB do not include a concentration-dependent correction to the proton reference energy, thus effectively assuming pH = 0, so that users can unambiguously add any desired correction to their analysis.[19] To a first approximation, accounting for pH involves changing the proton chemical potential; however, a more rigorous pH treatment is complex and involves explicit solvation,[32] which is not currently included in our methods. The database reaction energies and adsorption energies only include free energy corrections for the molecular reference state energies under the assumption that vibrational corrections to the free energy for surface states cancel when calculating energy differences.

For the adsorption reaction shown below, where an adsorbate, A^*, adsorbs to an open surface site, *, to form the bound A^* state, the grand-canonical adsorption energy, $\Phi_{ads}[\mu]$, is thus calculated with eq 2:

$$A + ^* \rightarrow A^*$$

$$\Phi_{ads}[\mu] = \Phi_{A*}[\mu] - (\Phi_{Surf}[\mu] + \Phi_A[\mu]) \tag{4}$$

using the grand free energies of the adsorbed state, clean surface, and the appropriate set of molecular reference states. Note that although the potential must be kept constant across all grand free energies in eq 2, the number of electrons used to calculate each grand free energy may differ within the grand-canonical ensemble. In addition to GC-DFT-computed adsorption energies, BEAST DB contains adsorption energies calculated without applied potential using DFT Helmholtz free energies for uncharged systems. These adsorption energies facilitate comparisons to the literature that has used the CHE approach. The electrochemical potential μ can be defined as an applied potential, U, relative to a particular reference electrode, U_{ref}, using eq 3:

$$\mu = -(U_{Ref} + U) \tag{5}$$

Although the reference electrode potential may be defined using the widely used standard hydrogen electrode value of 4.44 eV, in this work, we instead use U_{ref} = 4.66 eV, which is calibrated using the potentials of zero charge (PZC) of several transition metal surfaces for the CANDLE implicit solvation model.[26] Overall, these solvated, grand-canonical calculations capture the full, nonlinear variation of the grand free energies as a function of electrode potential, in contrast to the linear extrapolation of the grand free energy from neutral calculations in the conventional CHE approach.[19] We note that the above formalism was used to calculate the reaction pathways in BEAST DB as well. Additional details for these more complex calculations are described in detail at https://beastdb.nrel.gov/about.

2.2. GW Approximation and Random Phase Approximation. In this work and in BEAST DB in the future, GW calculations are performed by solving the Dyson equation for the quasiparticle (QP) energies and wave functions (using Hartree atomic units),[33]

$$\left[\frac{-\nabla^2}{2} + V_{ion} + V_H + \Sigma(E_{n\mathbf{k}}^{QP})\right]\psi_{nk}^{QP} = E_{n\mathbf{k}}^{QP}\psi_{n\mathbf{k}}^{QP} \tag{6}$$

where ∇ is the kinetic energy operator, V_{ion} is the ionic potential, V_H is the Hartree potential, Σ is the nonlocal, energy-dependent self-energy, and $E_{n\mathbf{k}}^{QP}$ and $\psi_{n\mathbf{k}}^{QP}$ are the quasiparticle energies and wave functions, respectively. For both GW and RPA calculations, we use the single-particle wave functions, ψ_{nk}^{KS}, from PBE DFT calculations as our quasiparticle wave functions (the diagonal approximation), along with the Kohn–Sham energies, $\varepsilon_{n\mathbf{k}}^{KS}$. Within the GW approximation, Σ depends on the one-body Green's function, G, and the dynamically screened Coulomb interaction, W. W depends on the full-frequency dielectric matrix, which in turn depends on the independent-electron polarizability, χ^0, as is standard[34] within the random phase approximation.[35] In the plane-wave basis, χ^0 can be written as

$$\chi^0_{\mathbf{GG'}}(\boldsymbol{q};\omega) = \sum_{\substack{n occ.\\ n' emp.}} \sum_{\boldsymbol{k}} \frac{2M^*_{nn'}(\boldsymbol{k}, \boldsymbol{q}, \mathbf{G})M_{nn'}(\boldsymbol{k}, \boldsymbol{q}, \mathbf{G}')}{\varepsilon_{n\mathbf{k}+\boldsymbol{q}} - \varepsilon_{n'\mathbf{k}} \pm \omega + i\delta} \tag{7}$$

with the matrix elements $M_{nn'}$ defined by

$$M_{nn'}(\boldsymbol{k}, \boldsymbol{q}, \mathbf{G}) \equiv \langle\psi_{n\mathbf{k}+\boldsymbol{q}}|e^{i(\boldsymbol{q}+\mathbf{G})\cdot r}|\psi_{n'\mathbf{k}}\rangle \tag{8}$$

Here, $\boldsymbol{q}$ is a vector in the first Brillouin zone, $\mathbf{G}$ and $\mathbf{G}'$ are reciprocal-lattice vectors, ω is the frequency, δ is an infinitesimal that locates the poles of χ^0 relative to the real frequency axis, and n and n' count occupied and empty bands, respectively. See refs 36–44 for detailed reviews of the GW and RPA methods.

The RPA correlation energy is computed from χ^0 within the adiabatic connection-fluctuation dissipation theorem (ACFDT) as

$$E_{corr}^{RPA} = \int_0^\infty \frac{d\omega}{2\pi} Tr\{\ln[1 - v\chi^0(i\omega) + v\chi^0(i\omega)]\} \tag{9}$$

where Tr denotes the trace of a matrix, and v is the bare Coulomb interaction. RPA total energies were calculated by subtracting the exchange correlation contribution to the DFT Helmholtz energy, E_{xc}, from the DFT energy and adding in RPA correlation energy and the exact exchange energy computed with the input PBE orbitals.

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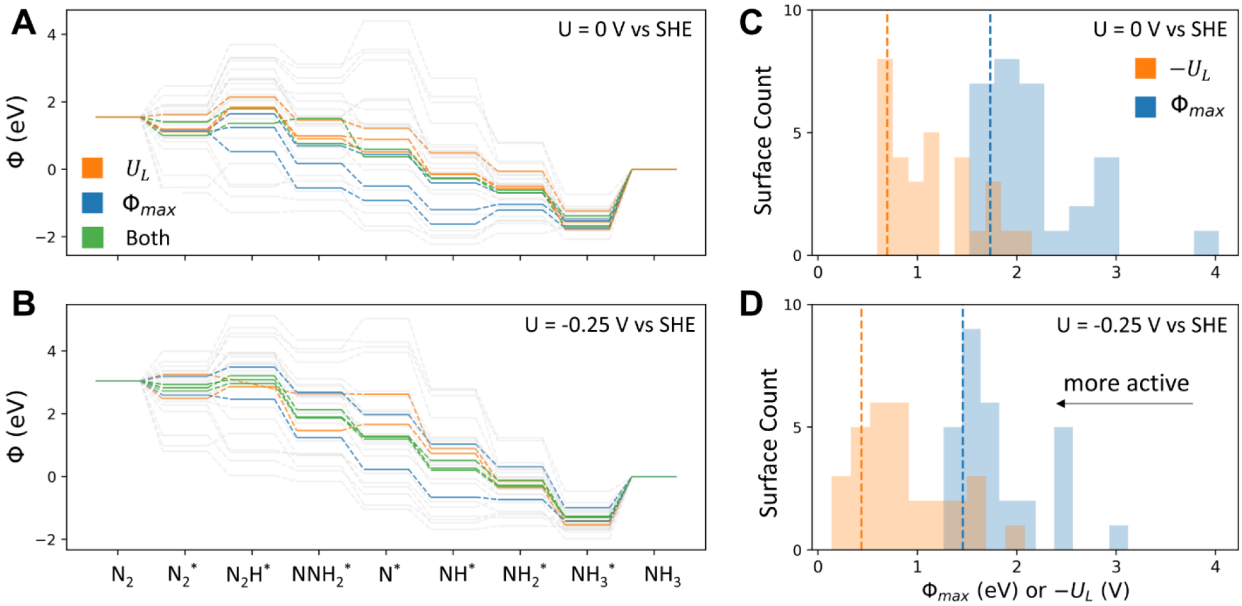

Figure 4. Thermodynamic descriptor selection on free energy pathways. Free energy diagrams at 0 and −0.25 V vs SHE are shown in panels (A) and (B), respectively. Panels (C) and (D) show Φ_{max} (blue) and $-U_L$ (orange) for all materials with a bin size of 0.25. Materials are sorted by the Φ_{max} and $-U_L$ descriptors, and a cutoff is drawn for each descriptor at the 5th most active material according to each descriptor. The cutoffs are drawn in panels (C) and (D) with dashed lines. Materials that fall within the Φ_{max} cutoff are drawn in blue in panels (A) and (B), and materials that fall within the $-U_L$ threshold are drawn in orange. Materials that fall within both descriptor thresholds, meaning that both $-U_L$ and Φ_{max} predict these materials to be active, are drawn in green. $-U_L$ and Φ_{max} produce qualitative differences in catalyst ranking, with $-U_L$ selecting more weakly binding catalysts relative to Φ_{max} because Φ_{max} penalizes catalysts that under-bind N_2* and N_2H* after desorbing the overbound NH_3* state.

2.3. High-Throughput Computational Workflow. Bulk structures were optimized in JDFTx, and surfaces were cut from the optimized structures using the pymatgen Python package surface generation functions. Surfaces were relaxed at constant potential and constant charge under the BEAST DB standard solvation parameters, which are described in detail at https://beastdb.nrel.gov/about. The pymatgen AdsorbateSiteFinder was used to enumerate unique active sites on the relaxed slabs. Adsorbate molecules were optimized in JDFTx under the standard solvation parameters before being placed on the adsorption sites of optimized surfaces. Adsorption calculations at a given potential were only initialized on surface structures that had been optimized at the same potential. Adsorbate structures were optimized, and binding energies were calculated from the optimized surface and adsorbate structures. The scripts used to run the calculations are hosted on our Github repository: https://github.com/Nick-Singstock/BEAST_DB_Manager. To standardize our workflows further and make external data contribution to BEAST DB possible, the next database milestone includes the refactoring of our workflows in FireWorks.[45]

3. RESULTS

3.1. Thermodynamic Activity Descriptors. Rapid evaluation and comparison of catalytic activity are important to aid researchers in selecting and designing performant catalysts. BEAST DB allows users to compare catalysts using multiple descriptors based on binding energies. The most widely used descriptor for electrocatalysts is the limiting potential (U_L),[19] which is defined as the minimum potential required to make all electrochemical steps thermoneutral or favorable. This descriptor is not only useful for ranking materials by predicted activity but also for estimating the potential at which the reaction proceeds. Our database facilitates screening of materials by limiting potential as shown in Figure 4.

The concept of limiting potential was initially defined for mechanisms comprised of purely PCET steps where the sensitivity of the step energy to potential is formally given by the change in the energy of the transferred electron.[19] However, GC-DFT facilitates the calculation of reaction steps that do not involve formal electron transfers (i.e., chemical steps), which are assumed to be insensitive to applied potential in the limiting potential scheme. The energetic span (Φ_{max})[46,47] allows for the evaluation of mechanisms that contain non-PCET steps that are still sensitive to potential (see Section 3.3 for a potential-sensitive non-PCET step), which is calculated following the methods of our previous paper that used Φ_{max} to assess the NRR scaling relations.[30] Figure 4A,B shows the calculated free energy diagrams for the NRR on flat and stepped monometallic surfaces. Colored lines rank within the top five most active catalysts predicted by the $-U_L$ (orange) and Φ_{max} (blue) descriptors, while gray lines are all other less active catalysts screened. Figure 4C,D shows the distribution of $-U_L$ and Φ_{max} across the monometallic data set. Materials are sorted by their U_L and Φ_{max} values, and the material with the fifth lowest $-U_L$ and Φ_{max} defines a cutoff that separates the five most active materials predicted by either descriptor from the rest of the materials. Those cutoffs are drawn as dashed lines in Figure 4C,D, and their corresponding free energy pathways are colored by the descriptors that rank them in the top five most active catalysts. Using these two methods to select materials leads to mostly different sets with some overlap (green lines in Figure 4A,B). Φ_{max} tends to select materials that bind adsorbates more strongly than the materials selected by U_L because Φ_{max} penalizes materials that under-bind N_2* and N_2H* after desorbing the overbound NH_3* state. In contrast, activity predictions based on U_L are

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Table 1. Data for Materials from Figure 4[a]

material	0 V vs SHE Φ_{max}	U_L	Φ_{max} rank	U_L rank	−0.25 V vs SHE Φ_{max}	U_L	Φ_{max} rank	U_L rank
Fe(110)	1.52	−1.03	1	15	1.41	−0.72	3	16
Ru(111)	1.56	−0.70	2	6	1.27	−0.31	1	3
Mo(110)	1.61	−1.44	3	23	1.52	−0.60	8	13
Co(111)	1.63	−0.61	4	2	1.34	−0.27	2	2
Rh(211)	1.68	−0.69	5	5	1.59	−0.56	12	11
Re(111)	1.73	−1.20	6	20	1.46	−0.74	6	17

[a] Φ_{max} is in units of eV and U_L in V vs SHE. The six materials with the lowest Φ_{max} are shown and ordered by Φ_{max} at 0 V. The rank of each material represents the sorting order of the material according to the specified thermodynamic descriptor at the specified potential, with a rank of 1 denoting the predicted best catalyst.

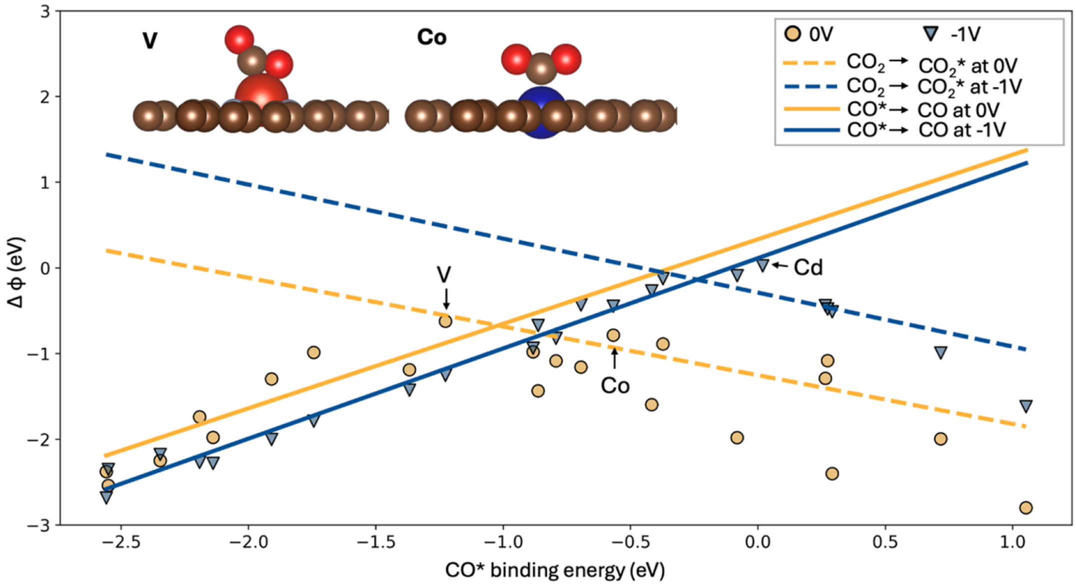

Figure 5. Theoretical volcano for CO2R on MN_4C. Scaling of the largest elementary step change in grand free energy ($\Delta\Phi$ of each catalyst with the binding energy of CO calculated at constant charge. The solid lines fit the CO desorption step to the constant charge CO binding energy, while the dashed lines fit the CO_2* adsorption step to the constant charge CO binding energy. Yellow and blue lines indicate 0 V vs SHE and −1 V vs SHE, respectively. The data points show the positions of various MNC catalysts on the volcano plot. Circles represent 0 V vs SHE data, and triangles represent −1 V vs SHE data. The CO_2 binding geometries on vanadium and cobalt MNCs are shown on the upper left of the plot. The MNC with a Co metal center is located near the top of the volcano, indicating its high predicted catalytic activity at 0 V relative to other catalysts.

unaffected by the energetics of the NH_3 desorption step because it does not involve a formal PCET. Because Φ_{max} takes account of more elementary steps than U_L, its material ranking is expected to match experimental trends more closely. Table 1 shows the thermodynamic activity descriptors and their limiting steps for the materials selected in Figure 4 at both potentials. The material ranking changes between biases with both descriptors, which shows that the reaction step energies are not all perfectly linear with the applied potential. Furthermore, materials predicted to be active by Φ_{max} are not necessarily predicted to be active by U_L, although there is some overlap in both descriptors across potentials such as Co(111) and Ru(111).

Both the limiting potential and energetic span descriptors enable the comparison of electrocatalysts across a material space as a starting point for designing novel catalysts and selecting catalysts for experimental evaluation. However, we note that if the kinetically limiting steps are understood to be purely chemical, the energetic span descriptor is likely a better predictor of catalytic activity.[30,46]

3.2. Scaling Relations. Scaling relations describe the linear correlation between the binding energies of reaction intermediates on the catalyst surface and provide valuable insights into the catalytic activity of materials.[48,49] These relations give rise to the characteristic volcano relationships frequently observed between intermediate binding energies and catalytic activity. BEAST DB facilitates comprehensive scaling relation analysis across various applied electrode potentials using reaction mechanisms that go beyond PCET mechanisms, allowing for the identification of limiting steps and their sensitivity to the applied bias. To this end, scaling relations for a given reaction and material class can be calculated from BEAST DB. Then, individual materials can be mapped onto the resulting volcano plot to assess their proximity to the volcano peak and identify thermodynamic limitations.

Here, we highlight the scaling relations for the CO_2 reduction reaction on MNC catalysts. These scaling relations rely on a hypothetical mechanism, which follows the path $CO_2 \rightarrow CO_2^* \rightarrow COOH^* \rightarrow CO^* \rightarrow CO$.[50] For CO_2R on MNC catalysts, the MNC with a Co metal center (Co-MNC) is found at the top of the volcano at 0 V vs SHE, indicating its high predicted catalytic activity (Figure 5). Co-MNC undergoes the hypothesized mechanism, as shown in the figure. This

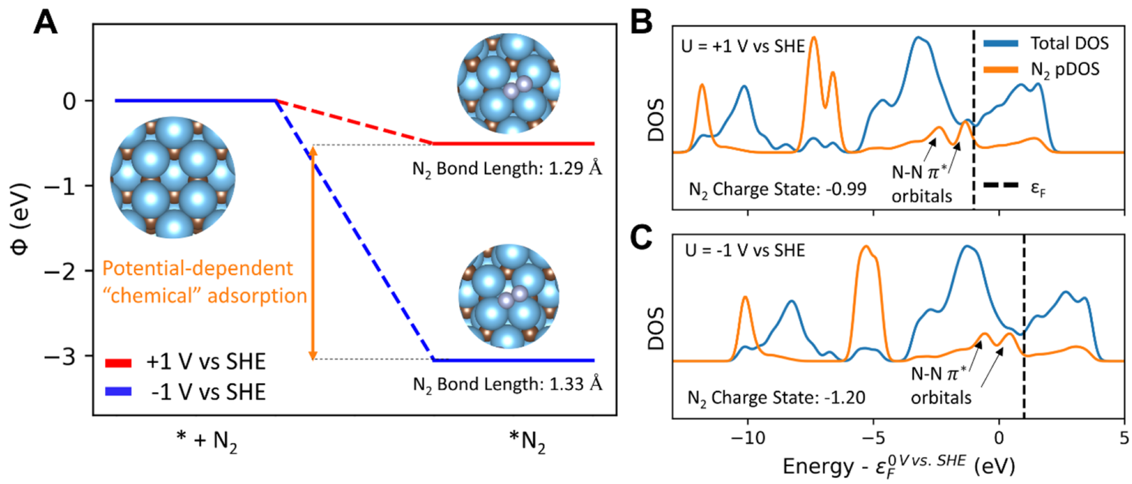

Figure 6. Bias-dependent N_2 adsorption. (Left) Adsorption of N_2 at 0 and −0.5 V vs SHE on TiC $(11\bar{1})$. Although the step contains no formal electron transfer, it is highly sensitive to bias. (Right) Total density of states (DOS) and projected density of states (pDOS) projected onto N atoms. The *x*-axis is zeroed at 0 V vs SHE, showing that the Fermi level changes on an absolute scale with the electrode potentials. The net bond order decreases as indicated by the increased N−N bond length, greater population of $N_2\pi^*$ orbitals, and increased charge localized on N_2.

computational prediction is consistent with experimental observations of MNC catalysts, which have also identified Co-MNC as one of the most active catalysts for CO_2R.[51] The position of Co-MNC near the yellow dashed scaling line suggests that its activity is limited by the CO desorption step, and selectively weakening the CO* binding energy could improve its performance.

However, not all surfaces will undergo this exact mechanism. For instance, our GC-DFT calculations reveal that the vanadium MNC (V-MNC) adsorbs CO_2 by bonding to both the carbon and one of the oxygens, suggesting that the scaling relations predicated on the assumed mechanism above are not predictive of the activity or limiting steps for V-MNC. This is evident in Figure 5, where V-MNC lies above the scaling line at 0 V, and thus, the scaling relations for our mechanism are not applicable. The figure excludes the MNCs that fail to adsorb CO_2. Moreover, Cd MNC is plotted for −1.00 V in Figure 5 because GC-DFT predicts that CO_2 will not adsorb at 0 V vs SHE but will adsorb at −1 V vs SHE.

When the applied bias becomes more reducing (more negative), the peak of the volcano shifts upward and to the right (Figure 5). The vertical shift is expected because a more reducing potential makes the overall reaction more favorable. However, the vertical shift is relatively small because the CO_2 adsorption and CO desorption steps are chemical steps that are less sensitive to the applied potential compared to electrochemical steps that include formal electron transfers. The more pronounced shift to the right indicates that at a more reducing applied potential, the optimal catalyst should have a weaker affinity for CO.

We note that if the CHE model were used instead of GC-DFT, plotting these scaling relations at multiple potentials would not be possible, as the CHE model cannot differentiate between the limiting scaling lines at 0.00 and −1.00 V vs SHE because they contain no formal electron transfers and the shift of the volcano peak at −1.00 V vs SHE would not be observed.

3.3. GC-DFT Electronic Structure Effects. The energetics of elementary electrochemical reactions are often well approximated by CHE, but steps that do not contain formal PCETs will not be impacted by the CHE correction. However, the energy change of such reactions is not necessarily independent of applied potential. One example is the adsorption of CO_2 to the catalyst surface, which is understood to be a reductive adsorption decoupled from any proton transfer.[50] Modeling this process with GC-DFT shows that the CO_2 adsorption step, which experimentalists have identified as the rate-determining step,[52] is highly sensitive to applied potential and may only adsorb at a given onset potential in some cases.[53] GC-DFT and the data in our database are appropriate tools for studying decoupled electron transfers but are also useful in understanding how the electronic structure changes with applied potential and how these changes manifest in reaction energetics.

For some surfaces reported in the data set, the electronic structure of the adsorbate−surface complex changes substantially as a function of applied potential, which manifests as a bias dependence of steps that are classically thought of as purely chemical. For example, the capacitance of chemical bonds may allow for significant electron transfer as the potential changes,[54] as in the N−N bond in N_2 adsorbed onto TiC $(11\bar{1})$, as is shown in Figure 6A. Its adsorption energy decreases by 2.54 eV when the potential is lowered from +1.00 to −1.00 V vs SHE. The π^* antibonding states in N_2 (two orange peaks labeled with arrows in Figure 6B) lie partially below the Fermi level at +1.00 V vs SHE, and as potential is made more reducing, the antibonding orbitals are further populated, causing the N−N bond length to increase from 1.29 to 1.33 Å. This potential-dependent transfer of electrons in part leads to a highly potential-sensitive adsorption energy despite the lack of a formal electron transfer in the adsorption step. Population of these orbitals may signify a propensity to dissociate N_2, which could result in a step change in reaction mechanism as the applied potential changes.

Changes in electronic structure that deviate from the rigid-band approximation are important in understanding the impact of applied potential on reactivity.[54] Recently, several methods have been proposed to approximate the effects of the applied potential through post hoc double layer charging and dipole-field interaction corrections.[55,56] While these approaches successfully describe electrochemical phenomena in many cases, they are incapable of capturing changes in electronic

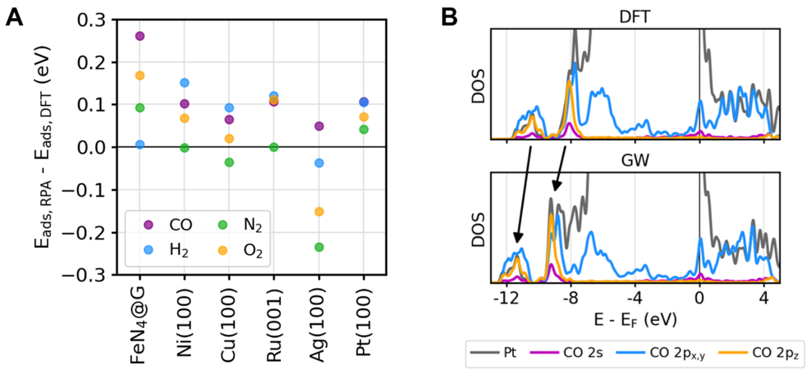

Figure 7. (A) Change in adsorption energy for CO, H_2, N_2, and O_2 adsorbed onto (1 × 1) Ag(100), Cu(100), Ni(100), Pt(100), Ru(001), and (3 × 3) FeN_4@G surfaces using RPA as compared to PBE. The molecules were adsorbed at full coverage on the transition metal surfaces. Positive values indicate destabilization using RPA. (B) GW projected density of states (DOS) for the Pt(100) + CO system as compared to the PBE DOS. The arrows highlight occupied bonding states that shift further in energy below the Fermi level using the GW approach.

structure that are self-consistently calculated within GC-DFT and any ionic rearrangements they may induce.

3.4. Features under Development. *3.4.1. Beyond-DFT Properties.* In addition to solvation and potential effects, we are investigating how the beyond-DFT GW/RPA approaches impact electrocatalytic system predictions and potentially correct issues related to semilocal methods. For RPA total energies, we implemented algorithmic improvements that reduce the system-size scaling to improve RPA accessibility for catalytically relevant systems.[31,57−59] For GW electronic structures, we established a formalism to more seamlessly connect the GW framework to nonlocal implicit solvation models and directly incorporate the effect of liquid screening.[60] The development of a framework to calculate RPA total energies from solvated GC-DFT calculations using applied potential is the subject of ongoing work. Once complete, these features will be integrated into BEAST DB and released for public use. The next release of BEAST DB, however, will include RPA corrections to the DFT adsorption energies for a selected set of catalysts in vacuum, i.e., not in the presence of solvation or applied potential.

We have used GW and RPA to screen small surface models and adsorbate combinations relevant to electrocatalysis (Table S2) and will be applying these techniques to larger surface models with more structurally diverse facets. The current set of GW electronic structures and RPA total energies includes CO, H_2, N_2, and O_2 adsorbed onto (1 × 1) Ag(100), Cu(100), Ni(100), Pt(100), Ru(001), and (3 × 3) FeN_4@G surfaces in vacuum at full coverage. These structures were generated from the same high-throughput DFT screening approach used to generate the structures for the rest of BEAST DB. Figure 7A shows that RPA destabilizes the adsorption energies of nearly all adsorbate/surface combinations relative to PBE calculations. Table S2 summarizes the RPA results and shows that they are in excellent agreement with the existing literature.[15] We find that CO and O_2 adsorbed to FeN_4@G are particularly destabilized, while N_2 and O_2 adsorption to Ag(100) is particularly stabilized. The corresponding GW projected density of states (DOS) shows that adsorption energy destabilization corresponds with the bonding orbitals of the adsorbates shifting to much lower energies relative to the Fermi level. An example of this for the Pt(100) + CO system is shown in Figure 7B. Although still in development, we believe that the incorporation of beyond-DFT methods will play a key role in further understanding the catalysts found from the above high-throughput screening approaches.

3.4.2. Machine Learning Beyond-DFT Effects. Although GW quasiparticle eigenvalues have been shown to more accurately reflect the electronic structure of materials than DFT eigenvalues, the higher cost of GW calculations relative to DFT calculations has historically hindered such high-throughput calculations. Knøsgaard and Thygesen have developed a machine learning approach for learning the GW band structures of 2D materials based on fingerprints of individual electronic states.[61] Ongoing work in our project will leverage BEAST DB to facilitate the development of ML models that Δ-learn GW quasiparticle eigenvalues from DFT eigenvalues. Efforts to create a large set of GW eigenvalues to Δ-learn are currently ongoing; however, we have previously established a baseline framework for the capability by training an ML model that accurately predicts hybrid functional HSE06 eigenvalues of diverse bulk materials using a small feature set including DFT eigenvalues, projectors, and simple charge transfer metrics.[62] As an example that is relevant to the performance of Mg-air batteries,[63] Figure 8 shows that the band structure for rock salt MgO predicted using the ML model significantly improves the semilocal calculated band energies relative to hybrid functionals. Our success in Δ-learning the HSE06 eigenvalues suggests that a similar approach could be used to predict GW eigenvalues. Furthermore, ongoing work is being performed to extend this approach to the prediction of a surface with adsorbate electronic structures using a graph neural network, which accounts for a local active site structure. We expect that this

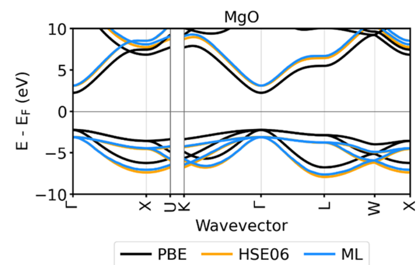

Figure 8. Machine-learned eigenvalues. Rock salt MgO band structure predicted using a semilocal functional (PBE; black), a hybrid functional (HSE06; orange), and ML (blue).

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will further accelerate accurate generation and analysis of surface electronic structures, which will in turn be used to better understand trends in catalytic activity at the RPA level, as discussed in Section 3.4.1.

4. DISCUSSION AND CONCLUSIONS

BEAST DB provides the community with a wealth of realistically computed electrochemical catalysis data and a set of tools to analyze the data. The database has continuing support and will be expanded significantly in the coming years to include more realistic surface models with variable coverages and coadsorptions of aqueous species, more electrochemical reaction pathways, and calculations using higher-fidelity methods. The catalytic surfaces presented here are clean surfaces with a low coverage of adsorbates (except the RPA-calculated adsorption energies) with no competing species. In reality, catalytic surfaces are usually defected and have a high coverage of various intermediates and electrolyte species, and the mechanisms on these surfaces can involve defects, e.g., lattice oxygen. Furthermore, in addition to reaction thermodynamics, kinetics are crucial to providing insight into catalytic activity and selectivity. Current methods for calculating reaction rates suffer from nonrobustness due to transition states being at a saddle point in contrast to an energy minimum. Additionally, both thermodynamics and kinetics are affected by the quality of the underlying theory used to describe the reaction energetics, and further work is needed to make beyond-DFT results more accessible to a larger range of reactions and chemistries. Finally, more advanced solvation models that capture an atomic-scale liquid structure using classical-density functional theory or integral equation theories would allow for a seamless description of the electrocatalytic interface, including double layers and inhomogeneity of the fluid dielectric and ionic response from the interface to the bulk electrolyte. Ongoing work of the BEAST team targets generalizing these methods, robust enough to be applied to large sets of materials, on par with the current applicability of implicit solvation models.

With the current capabilities of BEAST DB and the planned work described above, the potential benefits to the catalysis community are significant. The ability to look at different catalysts and how their reaction thermodynamics and kinetics change as a function of potential and with different representations of surfaces coverages and defects will be beneficial to both theoretical and experimental catalysis researchers. Additionally, access to the atom-projected charges and spins at the catalyst active site and on the bound molecule using a consistent methodology facilitates a comparison to experimental spectroscopic techniques such as XANES and for the rationalization of observed trends. The combination of the GW electronic structure and RPA reaction energetics with ML would enable an unprecedented level of accuracy in electrocatalytic calculations, with considerable benefits for the calculation of transition states and energetics in systems with localized electronic states. Access to a more complete description of electrocatalytic systems and more direct and transparent comparisons to experiments will enable the rational design of better electrocatalysts for a range of chemistries important to decarbonizing the world economy.

ASSOCIATED CONTENT

Data Availability Statement

All DFT data discussed in this study are available at https://beastdb.nrel.gov/. The results and methods used to perform the GW/RPA calculations in this study are presented in Table S2 and accompanying text of the Supporting Information. The ML data presented in Section 3.4.2 are available at https://github.com/Sutton-Research-Lab/ML_eigenshift. All DFT calculations were run using the open-source code, JDFTx, available at https://jdftx.org/. Calculation workflows were managed using our high-throughput workflow manager available at https://github.com/Nick-Singstock/BEAST_DB_Manager. GW/RPA calculations were performed using the open-source BerkeleyGW software package available at https://berkeleygw.org/download/.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcc.4c06826.

Additional GC-DFT and RPA validation and structure change analysis (PDF)

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Notes

The authors declare no competing financial interest.

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